

SPATIO-TEMPORAL MODELS WITH INLABRU

Noumea (New Caledonia) 10-14 March 2026

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1 Welcome to the course!

- Welcome to the `inlabru` workshop in New Caledonia!
- The aim of this workshop is to introduce you to a range of statistical modelling approaches, in particular the temporal, spatial and spatio-temporal modelling as implemented in the `inlabru` package.
- Workshop materials are available in the github repository

⚠ Important! □

Please use the instructions provided [here](#) to install and check your installation *before* the course start.

2 Learning Objectives for the workshop

At the end of the workshop, participants will have an understanding of:

- the motivation for and the challenges of analysing and modelling spatial data
- statistical models used to analyse spatial and spatio-temporal data
- the implementation of these models in the `inlabru` package
- how to independently analyse spatial data with `inlabru`

3 Intended audience

The workshop aims to cater for participants with a range of different backgrounds, who is interested in analysing data with modern spatial and spatio-temporal statistical modelling approaches.

4 Prerequisites

Participants should be familiar with the R environment, and general statistical approaches for modelling such as regression, analysis of (co)variance, and generalized linear models.

No knowledge of R-INLA or `inlabru` is required.

5 Schedule

NOTE The schedule is indicative and some changes might come!

5.0.1 Day 1

Time	Topic
10:00 - 10:30	Welcome
10:30 - 11:30	Session 1: Introduction to inlabru
11:30 - 13:00	Session 2: Some practical tips on inlabru
13:00 - 14:30	Lunch break ☐
14:30 - 15:30	Practical Session 1
15:30 - 15:45	Coffee Break ☐
15:45 - 16:45	Practical Session 2
16:45 - 17:00	wrap-up and outlook

5.0.2 Day 2

Time	Topic
9:00 - 10:00	Session 3: Time Series Models
10:00 - 10:30	Snack ☐
10:30 - 11:30	Session 3 cont.: Time Series Models
11:30 - 13:00	Practical Session 3
13:00 - 14:30	Lunch break ☐
14:30 - 15:30	Session 4: Introduction to Spatial Statistics
15:35 - 15:45	Coffee Break ☐
15:45 - 16:45	Practical Session 4
16:45 - 17:00	wrap-up and outlook

5.0.3 Day 3

Time	Topic
9:00 - 10:00	Session 5: Areal processes
10:00 - 10:30	Snack ☐
10:30 - 11:30	Session 6: Geostatistics
11:30 - 13:00	Practical Session 5
13:00 - 14:30	Lunch break ☐
14:30 - 15:30	Session 7: Point Processes
15:35 - 15:45	Coffee Break ☐
15:45 - 16:45	Practical Session 5 (continued)
16:45 - 17:00	wrap-up and outlook

5.0.4 Day 4

Time	Topic
9:00 - 10:00	Session 8: Spatiotemporal models
10:00 - 10:30	Snack ☐
10:30 - 11:30	Session 9: Model comparison and evaluation
11:30 - 13:00	Practical Session 6
13:00 - 14:30	Lunch break ☐

Time	Topic
14:30 - 15:30	Session 10: Multi-likelihood/joint likelihood models
15:35 - 15:45	Coffee Break ☐
15:45 - 16:45	Practical Session 7
16:45 - 17:00	Closing session